Polynomials without variables: Box Arithmetic and the Fundamental Identity of Arithmetic

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Abstract

Box Arithmetic is a new foundation for mathematics built with boxes. Starting from a single object — the empty box — natural numbers, polynumbers, and polynomials in several variables arise by successive levels of nesting, with a natural hierarchy of arithmetic operations, and with no need for a notion of variable. After addition and multiplication, the third operation in this hierarchy, the caret, gives a new algebraic structure to polynumbers that leads to the Fundamental Identity of Arithmetic: for any natural number n, a finite caret product of prime power boxes equals exactly the box $\lfloor 1,2,...,n\rfloor$.

This subsumes the Fundamental Theorem of Arithmetic and yields the standard algebra of Dirichlet series — Möbius inversion, Euler products, divisor identities — by purely finite combinatorial reasoning, with no analysis required.

1 Introduction

Four millennia ago, the Babylonians already knew how to work with what we would now call polynomial equations, solving quadratics by a kind of geometric completion. Since then, mathematicians have steadily built an enormous edifice on top of this ancient foundation — and in doing so have quietly accumulated a great deal of logical debt. Even something as elementary as a polynomial in one variable, in the modern ZFC framework, requires an infinite set of ordered pairs of natural numbers, where each ordered pair $[a,b]$ is itself defined as the set $\{\{a\},\{a,b\}\}$. But a polynomial like $3+x+2x^3$ is a finite expression. Its definition should not require infinity.

We would like to propose a leaner foundation, called *Box Arithmetic*, in which every object is genuinely finite, directly constructible, and immediately encodable on a computer. The primary data structure is not a set but a multiset — which we call a box — where repetition is allowed and order is unimportant. Starting from a single object, the empty box (which we call zero), every mathematical object in our framework is built by finite nesting. Natural numbers emerge at the first level, polynomials in one variable at the next, polynomials in arbitrarily many variables at the level after that. At each stage a new arithmetic operation appears, and it is the third of these — which we call the caret operation — that leads somewhere surprising.

Here is our destination. To a modern audience comfortable with infinite objects and infinite operations, we may write informally:

$$\lfloor 1,2,2^2,\cdots\rfloor \wedge \lfloor 1,3,3^2,\cdots\rfloor \wedge \lfloor 1,5,5^2,\cdots\rfloor \wedge \lfloor 1,7,7^2,\cdots\rfloor \wedge \cdots = \lfloor 1,2,3,4,5,6,7,\cdots\rfloor . \qquad \text{(FIA)}$$

On the left, we see one box of prime powers for each prime, all connected by the caret operation. On the right, we see the box of all positive integers. The equality of these two boxes, which are ultimately combinatorial objects, is the *Fundamental Identity of Arithmetic*, and the usual Fundamental Theorem of Arithmetic is an immediate consequence. With just a little more work, so is Euler's product formula for the zeta function, and in fact innumerable further identities involving the Möbius function, divisor counts, divisor sums, the Liouville function, and many more.

Now, the identity as written above is of course informal: both sides contain infinitely many elements, and the caret product on the left runs over infinitely many primes. This sits in apparent tension with our claim that everything in box arithmetic is finite. The resolution is that (FIA) is best understood not as a single identity between infinite objects, but as a *uniform family* of exact finite identities, one for each natural number n : for any n, the caret product of finitely many finite boxes of prime powers — one box per prime up to n, containing only those powers not exceeding n — equals exactly the finite box $\lfloor 1,2,\cdots,n\rfloor$.

The infinite version is a suggestive shorthand for this family, in the same way that a power series is shorthand for its sequence of partial sums. Every individual identity in the family involves only finite boxes and finitely many operations, and is directly checkable by computation. We will make this precise in Section 7 via a truncation operator.

The algebra of Dirichlet series from analytic number theory can then be developed in a purely algebraic and combinatorial fashion, with no complex numbers and no convergent series required. And even questions that classically demand real analysis — such as the limiting behavior of $\sum_{0<n\leq N}\frac{1}{\sigma(n)^3}$ as N grows — can be answered perfectly well by rational approximation intervals, improvable on demand by purely finite computation. This is, after all, what applied mathematics and modern computation actually deliver; and since real analysis ultimately grounds itself in equivalence classes of Cauchy sequences of rationals, an infinite construction built on an infinite construction, one may reasonably ask why we should not simply work with the finite rational approximations directly, and be honest about what we are computing.

In Box Arithmetic, the primary data structure, the box, is simply a finite multiset. This is a more natural starting point than a set: multisets can *always be added*, by combining their elements, independent of the nature of those elements. This operation — combining boxes — underlies almost all linear structure in mathematics. It is built into the foundation, not added later as an afterthought.

We develop the basic arithmetic of boxes in Section 2, establish natural numbers and polynumbers (our name for polynomials without variables) in Sections 3 and 4, extend to multinumbers (polynomials in arbitrarily many variables) in Section 5, and introduce the caret operation in Section 6. Section 7 sets up the summation operator needed for the number-theoretic applications, and Section 8 states and applies the Fundamental Identity of Arithmetic, culminating in a rigorous rational interval for a classical Dirichlet sum computed entirely by finite box arithmetic.

2 Boxes and their addition and multiplication

The **empty box** is an expression of the form $\lfloor \quad \rfloor$ consisting of a left lower square bracket, followed by a space, followed by a right lower square bracket. We call this empty box **zero**, and denote it with the **number** 0. Thus $0 \equiv \lfloor \quad \rfloor$. We declare that the empty box 0 is a box.

General boxes are defined recursively by starting with the empty box and then nesting boxes inside boxes, always with only a finite number of boxes involved. Crucially repetitions are allowed, so this is going beyond set theory.

The official definition is that if $A, B, \cdots, K$ are boxes, not necessarily distinct, then so is the expression $M = \lfloor A, B, \cdots, K \rfloor$, with the order of the boxes unimportant. The boxes $A, B, \cdots, K$ are called the **elements** of the box M. If the box X is an element of the box Y, then we write $X \in Y$. Boxes are **equal** if they contain exactly the same elements, possibly rearranged.

Example 1 *Since* 0 *is a box, we have that* $\lfloor 0 \rfloor$ *is a box: we call this* ***one****, and denote it by the number* 1. *The box* ***two*** *is* $2 \equiv \lfloor 0, 0 \rfloor$ *and the box* ***three*** *is* $3 \equiv \lfloor 0, 0, 0 \rfloor$. *Here is a more complicated box, showing that our naming of basic boxes allows different representations of the same box.*

$$p \equiv \lfloor 0, \lfloor \lfloor 0 \rfloor, \lfloor 0 \rfloor \rfloor, \lfloor 0, 0, 0 \rfloor, \lfloor 0, 0, 0 \rfloor \rfloor = \lfloor 0, \lfloor 1, 1 \rfloor, \lfloor 0, 0, 0 \rfloor, \lfloor 0, 0, 0 \rfloor \rfloor = \lfloor 0, [1, 1], 3, 3 \rfloor.$$

The fundamental operation on boxes is **addition**. If $A, B, \cdots, K$ are boxes, then the **sum** $A + B + \cdots + K$ is the box consisting of all the elements of the boxes $A, B, \cdots, K$. So for example if $M = \lfloor a, a, K \rfloor$ and $N = \lfloor a, \blacktriangle \rfloor$ and $P = \lfloor K, \Diamond \rfloor$ are boxes, with $a, K, \blacktriangle$ and $\Diamond$ previously defined boxes, then $M + N + P = \lfloor a, a, K, a, \blacktriangle, K, \Diamond \rfloor = \lfloor a, a, a, K, K, \blacktriangle, \Diamond \rfloor$. Addition of boxes is the mathematical equivalent of dumping all the contents of several physical boxes into a single new physical box. The simplest example is just that $0 + 0 = 0$: the sum of two empty boxes is an empty box.

Note that addition is not intrinsically a binary operation, for example

$$\lfloor 0, 0 \rfloor + \lfloor \lfloor 0, 0 \rfloor, 0 \rfloor + \lfloor \lfloor 0, 0, 0 \rfloor, \lfloor 0 \rfloor \rfloor = \lfloor 0, 0, \lfloor 0, 0 \rfloor, 0, \lfloor 0, 0, 0 \rfloor, \lfloor 0 \rfloor \rfloor = \lfloor 0, 0, 0, 1, 2, 3 \rfloor.$$

Addition of boxes is **commutative**, meaning that $A + B = B + A$ for any boxes A and B. It is **associative**, meaning that $(A + B) + C = A + (B + C)$ for any boxes A, B and C. It **supports cancellation**, meaning that if A, B and C are boxes, and $A + B = A + C$, then $B = C$. The empty box 0 is the **additive identity**, so that $0 + A = A + 0 = A$ for any box A.

We also note a more subtle **zero sum property**: for boxes $A, B, \cdots, K$, if $A + B + \cdots + K = 0$, then we must have $A = B = \cdots = K = 0$. Multiplication is the natural **successor** to the addition operation, and holds for all boxes. If $A, B, \cdots, K$ are boxes, then the **product** $A \times B \times \cdots \times K$ is the box consisting of all possible sums of the elements of the boxes $A, B, \cdots, K$, that is

$$A \times B \times \cdots \times K \equiv \lfloor a + b + \cdots + k : a \in A, b \in B, \cdots, k \in K \rfloor.$$

In case one or more of $A, B, \cdots, K$ equals 0, the product has no elements in it, so is 0.

The meaning of this notation is the same as in usual set theory, where the colon ":" is to be read as "such that" and what precedes the colon is an object which is going to be an element of the box, and what follows it is a condition or conditions on that object. However here multiplicities are maintained, so that we must include all possibilities, without removing duplicates.

So for example if $a, K, \blacktriangle$ and $\Diamond$ are boxes, and $M = \lfloor a, a, K \rfloor$ and $N = \lfloor a, \blacktriangle \rfloor$ and $P = \lfloor K, \Diamond \rfloor$, then

$$M \times N \times P = \left\lfloor \begin{array}{c} a+a+K, a+a+\Diamond, a+\blacktriangle+K, a+\blacktriangle+\Diamond, a+a+K, a+a+\Diamond, a+\blacktriangle+K, a+\blacktriangle+\Diamond, \\ K+a+K, K+a+\Diamond, K+\blacktriangle+K, K+\blacktriangle+\Diamond \end{array} \right\rfloor .$$

It follows that multiplication is not intrinsically a binary operation. Multiplication of boxes is however commutative, so that $A \times B = B \times A$ for any boxes A and B. It is also associative, so that $(A \times B) \times C = A \times (B \times C)$ for any boxes A, B and C. Multiplication **distributes over addition**, meaning that for any boxes A, B and C we have $A \times (B + C) = (A \times B) + (A \times C) = A \times B + A \times C$, where we adopt the usual convention that multiplication takes precedence over addition in the absence of brackets. The box $1 \equiv \lfloor 0 \rfloor = \lfloor 0 \rfloor$ is the **multiplicative identity**, so that $1 \times A = A \times 1 = A$ for any box A.

Multiplication supports **non-zero cancellation**, meaning that if A, B and C are boxes, with $A \neq 0$, and $A \times B = A \times C$, then $B = C$. There is also an important **zero product property**: for boxes $A, B, \cdots, K$, if $A \times B \times \cdots \times K = 0$, then we must have at least one of $A, B, \cdots, K$ equal to 0.

There is a natural hierarchy of operations on boxes, starting with addition, which we may write as $+ \equiv \times^{(0)}$, and then multiplication $\times \equiv \times^{(1)}$, and where at each stage the **successor** of the operation $\times^{(n)}$ is the operation $\times^{(n+1)}$ defined by

$$A \times^{(n+1)} B \times^{(n+1)} \cdots \times^{(n+1)} K \equiv \left\lfloor a \times^{(n)} b \times^{(n)} \cdots \times^{(n)} k : a \in A, b \in B, \cdots, k \in K \right\rfloor .$$

This is the box formed by all possible products $a \times^{(n)} b \times^{(n)} \cdots \times^{(n)} k$ where $a, b, \cdots, k$ are respectively from the boxes $A, B, \cdots, K$. We call the successor of multiplication the **caret** operation, and denote it by $\wedge \equiv \times^{(2)}$. We will see that it plays a crucial role in understanding number theory from a foundational point of view. Higher operations are not so easy to appreciate quickly, and they lead to a largely unfamiliar world of arithmetic.

For us a **type** T is a well-defined condition on a previously defined mathematical object that can be implemented in a computationally explicit way. We will adopt the convention that "a mathematical object O is of type T" will be expressed as $O \propto \mathrm{T}$. We will not regard a type as a mathematical object; it is rather a *property* of a mathematical object, or a *condition* satisfied by one.

3 Natural number arithmetic

Now we establish the starting point of arithmetic, after having introduced our fundamental notion of an empty box, or zero. A **natural number** is a box of zeros. The type of a natural number will be denoted by Nat. Thus $0 = \lfloor \quad \rfloor$ is a natural number (it is the empty box of zeros), and so we may write $0 \propto \mathrm{Nat}$. So is $1 \equiv \lfloor 0 \rfloor$ and $2 \equiv \lfloor 0, 0 \rfloor$ and $3 \equiv \lfloor 0, 0, 0 \rfloor$ and so on, using the usual Hindu Arabic notations, so that for example $14 = \lfloor 0,0,0,0,0,0,0,0,0,0,0,0,0,0 \rfloor$. The natural numbers are then **linearly ordered**, so up to a natural number n we have

$$0 < 1 < 2 < \cdots < n.$$

This ordering allows us naturally to define the **minimum** $\min(m,n)$ of natural numbers m and n, along with the **maximum** $\max(m,n)$. So $\min(7,3)=3$ while $\max(7,3)=7$.

The addition of natural numbers then becomes the usual familiar operation, for example $3+4 = \lfloor 0,0,0\rfloor + \lfloor 0,0,0,0\rfloor = \lfloor 0,0,0,0,0,0,0\rfloor = 7$. The linear ordering of natural numbers up to a natural number n can be described as repeated addition of 1, starting with 0 until n is reached.

When we restrict to boxes of zeros, we recover the usual multiplication of natural numbers. So to multiply 3 and 4, we form the box of all possible sums between elements of the boxes: $3\times 4 = \lfloor 0,0,0\rfloor \times \lfloor 0,0,0,0\rfloor = \lfloor 0,0,0,0,0,0,0,0,0,0,0,0\rfloor = 12$.

Here is an example of taking the product of several natural numbers, by taking all possible sums containing one element from each number:

$$2\times 2\times 1 = \lfloor 0,0\rfloor \times \lfloor 0,0\rfloor \times \lfloor 0\rfloor = \lfloor 0+0+0, 0+0+0, 0+0+0, 0+0+0\rfloor = \lfloor 0,0,0,0\rfloor = 4.$$

The arithmetic that we obtain of natural numbers, involving addition and multiplication, is just like the usual one in terms of computations, except that the foundation is very different. Addition and multiplication are defined for all boxes, so natural numbers allow us to introduce **multiples**: for a box A, we define $2A \equiv A+A = 2\times A$ and $3A \equiv A+A+A = 3\times A$, and in general $kA = A+A+\cdots+A = k\times A$ with k terms. For consistency we define $0A \equiv 0\times A = 0$ for any box A. We then have the general **multiple laws** that for any natural numbers m and n and any box A

$$mA + nA = (m+n)\,A \qquad \text{and} \qquad n\,(mA) = (m\times n)\,A.$$

Similarly we introduce **powers:** for a box A, we define $A^2 \equiv A\times A$ and $A^3 \equiv A\times A\times A$, and in general $A^k = A\times A\times\cdots\times A$ with k terms. For consistency we define $A^0 = 1$ for any box A. This means in particular that we also agree that $0^0 = 1$. We then have the general **power laws** that for any natural numbers k and m and any box A

$$A^k \times A^m = A^{k+m} \qquad \text{and} \qquad \left(A^k\right)^m = A^{k\times m}.$$

Furthermore if A and B are boxes, and n is a natural number, then $(A\times B)^n = A^n \times B^n$, and this generalizes in the obvious way to more boxes, so that if $A, B, \cdots, K$ are boxes, then

$$(A\times B\times\cdots\times K)^n = A^n\times B^n\times\cdots\times K^n.$$

If A and B are boxes, then the **multiplicity** $m_B(A)$ of B in A is the number of times B occurs as an element of A.

Example 2 *If* $A \equiv \lfloor 7,7,\lfloor 3,5\rfloor, \lfloor\lfloor 4\rfloor, 2\rfloor\rfloor$ *and* $B\equiv 7$ *then* $m_B(A) = 2$. *Similarly if* $C \equiv \lfloor 3,5\rfloor$ *then* $m_C(A) = 1$. *While if* $D = \lfloor 8\rfloor$ *then* $m_D(A) = 0$.

By recording multiplicities we may express a box more simply; we do that with a *left subscript* denoting the multiplicity. We can rewrite the box A in the example above as $A = \lfloor {}_2 7, {}_1\lfloor 3,5\rfloor, {}_1\lfloor\lfloor 4\rfloor, 2\rfloor\rfloor$, or simply $A = \lfloor {}_2 7, \lfloor 3,5\rfloor, \lfloor\lfloor 4\rfloor, 2\rfloor\rfloor$. The statement that 7 has multiplicity 2 in A can be rephrased as: the box 7 **occurs two times** in the box A.

If a box C has each of its elements occurring with multiplicity less than or equal to the multiplicity of that element in a box D, then we write $C \subseteq D$ and say that C is a **subbox** of D. The empty box 0 is a subbox of any box C, and also C itself is a subbox of itself.

Example 3 *The box* $\lfloor 0, 5, 5, 17 \rfloor = \lfloor 0,_2 5, 17 \rfloor$ *is a subbox of* $\lfloor 0, 0, 3, 5, 5, 17 \rfloor = \lfloor _2 0, 3,_2 5, 17 \rfloor$, *so that* $\lfloor 0, 5, 5, 17 \rfloor \subseteq \lfloor 0, 0, 3, 5, 5, 17 \rfloor$.

If a box C occurs as an element of A n times, and occurs as an element of B m times, then it will occur in $A + B$ a total of $m + n$ times. So we can then perform addition of general boxes in terms of addition of (natural number) multiplicities. So for example if $X \equiv \lfloor _8 0,_3 2,_2 \lfloor \lfloor \lfloor 4 \rfloor \rfloor \rfloor \rfloor$ and $Y \equiv \lfloor _3 0,_7 1,_5 2,_1 \lfloor \lfloor \lfloor 4 \rfloor \rfloor \rfloor,_9 \lfloor 8 \rfloor \rfloor$ then $X + Y = \lfloor _{11} 0,_7 1,_8 2,_3 \lfloor \lfloor \lfloor 4 \rfloor \rfloor \rfloor,_9 \lfloor 8 \rfloor \rfloor$.

We may also multiply boxes by working with multiplicities, such as

$$\lfloor _n A,_m B \rfloor \times \lfloor _p C,_q D \rfloor = \lfloor _{n \times p} (A + C),_{n \times q} (A + D),_{m \times p} (B + C),_{m \times q} (B + D) \rfloor .$$

Here both addition and multiplication of natural numbers are generally required.

Once we have natural numbers, we may define the **size** $s(M)$ of a box M to be the number of elements in M. Another way of saying this is that $s(M)$ is the box obtained by replacing each element of M with 0. Then clearly $s(A + B) = s(A) + s(B)$ and $s(A \times B) = s(A) \times s(B)$ and indeed $s\left(A \times^{(n)} B\right) = s(A) \times s(B)$ for any operation $\times^{(n)}$ in the hierarchy of arithmetical operations.

4 Polynumber algebra

We now introduce polynomial arithmetic, but our terminology is slightly different from usual, and the fundamental "variable" α has a very specific meaning.

A **polynumber** is a box of natural numbers. The type of a polynumber will be denoted by Poly. The **degree** $\deg(p)$ of a polynumber p is the largest natural number that appears as an element of it. So $p \equiv \lfloor 0, 2 \rfloor$, $q \equiv \lfloor 3, 5, 3 \rfloor$ and $r \equiv \lfloor 0, 0, 6, 0, 1, 3, 3 \rfloor$ are polynumbers,with $\deg(p) = 2$, $\deg(q) = 5$ and $\deg(r) = 6$.

Addition of polynumbers is just addition of boxes, so that for example with p and q as above, $p + q = \lfloor 0, 2, 3, 3, 5 \rfloor$. **Multiplication of polynumbers** is just multiplication of boxes, so that for example $p \times q = \lfloor 0, 2 \rfloor \times \lfloor 3, 5, 3 \rfloor = \lfloor 0 + 3, 0 + 5, 0 + 3, 2 + 3, 2 + 5, 2 + 3 \rfloor = \lfloor 3, 5, 3, 5, 7, 5 \rfloor$.

We introduce the simplest polynumber which is not actually a natural number, called **alpha:** $\alpha \equiv \lfloor 1 \rfloor$. Then $\alpha^2 \equiv \alpha \times \alpha = \lfloor 2 \rfloor$, $\alpha^3 = \lfloor 3 \rfloor$ and in general $\alpha^n = \lfloor n \rfloor$, and this allows any polynumber to be written in **polynomial form** as a sum of multiples of powers of α, so that for example $p = \lfloor 0, 2 \rfloor = \lfloor 0, 2 \rfloor = 1 + \alpha^2$ and $r = \lfloor 3, 3, 5 \rfloor = 2\alpha^3 + \alpha^5$ and $s = \lfloor 0, 0, 0, 1, 3, 3, 6 \rfloor = \lfloor 0, 0, 0 \rfloor + \lfloor 1 \rfloor + \lfloor 3, 3 \rfloor + \lfloor 6 \rfloor = 3 + \alpha + 2\alpha^2 + \alpha^6$.

Then the addition and multiplication of polynumbers agrees with the usual operations on polynomials, so that $p + q = \left(1 + \alpha^2\right) + \left(2\alpha^3 + \alpha^5\right) = 1 + \alpha^2 + 2\alpha^3 + \alpha^5$ and $p \times q = \left(1 + \alpha^2\right) \times \left(2\alpha^3 + \alpha^5\right) = \lfloor 3, 3, 5, 5, 5, 7 \rfloor = 2\alpha^3 + 3\alpha^5 + \alpha^7$.

There is another convenient way to represent a polynumber $p = m_0 + m_1\alpha + \cdots + m_n\alpha^n$, by introducing the **(linear) multiplicity array**

$$p = [m_0, m_1, \cdots, m_n]$$

using square brackets, that records the multiplicities of the natural numbers $0, 1, 2, \cdots, n$ appearing in p, in that order, up to a number $n \geq \deg(p)$. By allowing also additional zeroes in this way, this multiplicity array is not unique, so for example we could write $p = \lfloor 0, 2 \rfloor = 1 + \alpha^2 = [1, 0, 2] = [1, 0, 2, 0]$ and $r = \lfloor 3, 3, 5 \rfloor = 2\alpha^3 + \alpha^5 = [0, 0, 0, 2, 0, 1]$ and $s = \lfloor 0, 0, 0, 1, 3, 3, 6 \rfloor = 3 + \alpha + 2\alpha^2 + \alpha^6 = [3, 1, 0, 2, 0, 0, 1] = [3, 1, 0, 2, 0, 0, 1, 0, 0]$. Then addition of polynumbers corresponds to pointwise addition of multiplicity arrays, so that $p + r = [1, 0, 2] + [0, 0, 0, 2, 0, 1] = [1, 0, 2, 2, 0, 1]$.

Multiplication of polynumbers then corresponds to the **convolution**, or **Cauchy product** of multiplicity arrays. If $p = [p_0, p_1, \cdots, p_n]$ and $q = [q_0, q_1, \cdots, q_n]$ are polynumbers with possibly additional zeroes added, so that both arrays go up to a natural number n, then

$$p \times q = [p_0 q_0, p_0 q_1 + p_1 q_0, p_0 q_2 + p_1 q_1 + p_2 q_0, \cdots, p_n q_n]$$

ending in the $2n$-th position, where the k-th entry for $0 \leq k \leq 2n$ is

$$p_0 q_k + p_1 q_{k-1} + \ldots + p_k q_0.$$

Thus $p \times r = [1, 0, 2] \times [0, 0, 0, 2, 0, 1] = [0, 0, 0, 2, 0, 3, 0, 1]$. There is a convenient device to make the multiplication in this form easier, involving taking two strips of paper, with the multiplicity array of say p on one, and the reversed multiplicity array of q on the other, and then sliding one past the other sequentially, taking the sum of all products formed by adjacent entries.

Linear combinations of polynumbers are polynumbers. Power linear combinations of polynumbers are also polynumbers. With these conventions the algebra of polynumbers becomes the same as the familiar algebra of polynomials. But the difference now is that α is not an abstract variable, but just a particularly simple polynumber.

The **evaluation** of the polynumber $p = p_0 + p_1\alpha + p_2\alpha^2 + \ldots \ + p_n\alpha^n$ on the box A is the box

$$p(A) \equiv p_0 + p_1 A + p_2 A^2 + \ldots + p_n A^n.$$

For polynumbers p and q, a natural number λ and a box A, we have $(p + q)(A) = p(A) + q(A)$ and $(\lambda p)(A) = \lambda p(A)$ and $(p \times q)(A) = p(A) \times q(A)$. In particular if we evaluate a polynumber p on a natural number n, then the result $p(n)$ is also a natural number. There is another important special case. If $p = p_0\alpha^0 + p_1\alpha^1 + p_2\alpha^2 + \ldots \ + p_n\alpha^n$ is a polynumber and q is another polynumber, then the **composition** is the polynumber

$$p \circ q \equiv p(q) = p_0 + p_1 q + p_2 q^2 + \ldots \ + p_n q^n.$$

Note that because we do not have integers at this point, a factor theorem for polynumbers is not available, although we could still discuss divisibility and factors. However the Binomial theorem is directly accessible in terms of writing expansions for $(1 + \alpha)^n$ for a natural number n.

4.1 Truncations of polynumbers

If k is a natural number and p is a polynumber, then define the **degree** k **truncation of** p to be the polynumber $T_k(p)$ obtained from p by removing all elements which are strictly greater than k. Thus if $p = \lfloor 0,0,2,2,2,5,6,6 \rfloor = 2 + 3\alpha^2 + \alpha^5 + 2\alpha^6$ then $T_3(p) = \lfloor 0,0,2,2,2 \rfloor = 2 + 3\alpha^2$.

Then for polynumbers p and q we have $T_k(p+q) = T_k(p) + T_k(q)$ and also

$$T_k(p \times q) = T_k(T_k(p) \times T_k(q)).$$

This last identity is particularly important and reflects the property that the product of two polynumbers up to degree k depends only on the truncations of those polynumbers to degree k : higher degree coefficients have no effect.

Example 4 *Thus if p is as above and $q = \lfloor 0,3,3,4 \rfloor = 1 + 2\alpha^3 + \alpha^4$ then*

$$p \times q = \left(2 + 3\alpha^2 + \alpha^5 + 2\alpha^6\right) \times \left(1 + 2\alpha^3 + \alpha^4\right) = 2 + 3\alpha^2 + 4\alpha^3 + 2\alpha^4 + 7\alpha^5 + 5\alpha^6 + 2\alpha^8 + 5\alpha^9 + 2\alpha^{10}$$

so that $T_3(p \times q) = 2 + 3\alpha^2 + 4\alpha^3$ while $T_3(p) \times T_3(q) = \left(2 + 3\alpha^2\right) \times \left(1 + 2\alpha^3\right) = 2 + 3\alpha^2 + 4\alpha^3 + 6\alpha^5$ and indeed $T_3(T_3(p) \times T_3(q)) = 2 + 3\alpha^2 + 4\alpha^3$ also.

In fact this holds also for higher products: for any natural numbers n and k and polynumbers p and q one can show that

$$T_k\left(p \times^{(n)} q\right) = T_k\left(T_k(p) \times^{(n)} T_k(q)\right).$$

For polynumbers p and q, and a natural number k, we introduce the notion of **degree** k **equality** by declaring that

$$p =_k q$$

precisely when $T_k(p) = T_k(q)$.

5 Multinumbers

Now we go down one more level. A **multinumber** is a box of polynumbers. The type of a multinumber will be denoted by Multi.

Since natural numbers are polynumbers, any polynumber is also a multinumber. The multinumber $A = \lfloor 0,4,4,\lfloor 3,5 \rfloor,\lfloor 0,7,7,6 \rfloor \rfloor$ has the 5 elements $0,4,4,\lfloor 3,5 \rfloor$ and $\lfloor 0,7,7,6 \rfloor$, each of which is a polynumber, specifically $\lfloor 3,5 \rfloor = \alpha^3 + \alpha^5$ and $\lfloor 0,7,7,6 \rfloor = 1 + \alpha^6 + 2\alpha^7$. Thus we could rewrite $A = \left\lfloor 0,4,4,\alpha^3 + \alpha^5, 1 + \alpha^6 + 2\alpha^7 \right\rfloor$.

Multinumbers are closed under addition. They are also closed under multiplication, because polynumbers are closed under addition. So for example if $B = [2, 1+\alpha]$ and $C = \left\lfloor 5, \alpha^2, 1+\alpha \right\rfloor$ then $B + C = \left\lfloor 2, 1+\alpha, 5, \alpha^2, 1+\alpha \right\rfloor$ while $B \times C = \left\lfloor 7, 3+\alpha^2, 3+\alpha, 6+\alpha, 1+\alpha+\alpha^2, 2+2\alpha \right\rfloor$.

The simplest new multinumbers are the **base multinumbers** defined for a natural number $n \geq 1$ by

$$\alpha_n \equiv \lfloor \lfloor n \rfloor \rfloor = \lfloor \alpha^n \rfloor.$$

For consistency we may rewrite α as $\alpha_0 \equiv \alpha = \lfloor\lfloor 0\rfloor\rfloor = \lfloor 1\rfloor$. Then multinumbers give us polynomial algebra based on addition and multiplication of the base multinumbers $\alpha_0, \alpha_1, \alpha_2, \cdots \alpha_n$ for any natural number n. Let us see why.

Example 5 *The multinumber A from a previous example may be written as*

$$\begin{aligned} A &= \lfloor 0,4,4,\lfloor 3,5\rfloor,\lfloor 0,7,7,6\rfloor\rfloor = \lfloor 0\rfloor + \lfloor 4\rfloor + \lfloor 4\rfloor + \lfloor\lfloor 3\rfloor\rfloor \times \lfloor\lfloor 5\rfloor\rfloor + \lfloor\lfloor 0\rfloor\rfloor \times \lfloor\lfloor 7\rfloor\rfloor \times \lfloor\lfloor 7\rfloor\rfloor \times \lfloor\lfloor 6\rfloor\rfloor \\ &= 1 + 2\alpha_0^4 + \alpha_3\alpha_5 + \alpha_0\alpha_6\alpha_7^2. \end{aligned}$$

We see that the polynumber $\lfloor 3,5\rfloor = \alpha^3 + \alpha^5$ corresponds to the monomial term $\alpha_3\alpha_5$ while the polynumber $\lfloor 0,7,7,6\rfloor = 1 + \alpha^6 + 2\alpha^7$ corresponds to the monomial term $\alpha_0\alpha_6\alpha_7^2$.

For the polynumber $p \equiv k\alpha^m$ for natural numbers k and m, the corresponding multinumber is

$$\lfloor p\rfloor = \lfloor\lfloor m\rfloor + \lfloor m\rfloor + \cdots + \lfloor m\rfloor\rfloor = \lfloor\lfloor m\rfloor\rfloor \times \lfloor\lfloor m\rfloor\rfloor \times \cdots \times \lfloor\lfloor m\rfloor\rfloor = \alpha_m^k$$

with k copies in both the sum and the product. It follows that if q is the polynumber $q \equiv q_0 + q_1\alpha + q_2\alpha^2 + \cdots + q_n\alpha^n$ for some natural number n and natural numbers $q_0, q_1, \cdots, q_n$, then the multinumber $Q = \lfloor q\rfloor$ can be written as

$$\begin{aligned} Q &= \lfloor q_0 + q_1\alpha + q_2\alpha^2 + \cdots + q_n\alpha^n\rfloor = \lfloor q_0\rfloor \times \lfloor q_1\alpha\rfloor \times \cdots \times \lfloor q_n\alpha^n\rfloor \\ &= \alpha_0^{q_0} \times \alpha_1^{q_1} \times \cdots \times \alpha_n^{q_n} = \alpha_0^{q_0}\alpha_1^{q_1}\cdots\alpha_n^{q_n}. \end{aligned}$$

We say that the **degree** of Q is $q_0 + q_1 + \cdots + q_n$, and if $q_n \neq 0$ then the **extent** of Q is n.

A general multinumber M will be a sum of such terms, one for each of the polynumber elements of M. We may call this the **multinomial form** of M. The operations of addition and multiplication of multinumbers then correspond to the familiar operations on multinomials in the variables $\alpha_0, \alpha_1, \alpha_2, \cdots, \alpha_n$ for some natural number n.

Example 6 *If $B = \lfloor 0,\lfloor 3\rfloor,\lfloor 2,4\rfloor\rfloor = 1 + \alpha_3 + \alpha_2\alpha_4$ and $C = \lfloor\lfloor 1,1\rfloor,\lfloor 2,4\rfloor\rfloor = \alpha_1^2 + \alpha_2\alpha_4$ then*

$$B + C = \lfloor 0,\lfloor 3\rfloor,\lfloor 2,4\rfloor\rfloor + \lfloor\lfloor 1,1\rfloor,\lfloor 2,4\rfloor\rfloor = \lfloor 0,\lfloor 3\rfloor,\lfloor 1,1\rfloor,\lfloor 2,4\rfloor,\lfloor 2,4\rfloor\rfloor = 1 + \alpha_1^2 + \alpha_3 + 2\alpha_2\alpha_4$$

which agrees with $(1 + \alpha_3 + \alpha_2\alpha_4) + \left(\alpha_1^2 + \alpha_2\alpha_4\right) = 1 + \alpha_1^2 + \alpha_3 + 2\alpha_2\alpha_4$. Similarly

$$\begin{aligned} B \times C &= \lfloor 0,\lfloor 3\rfloor,\lfloor 2,4\rfloor\rfloor \times \lfloor\lfloor 1,1\rfloor,\lfloor 2,4\rfloor\rfloor \\ &= \lfloor 0 + \lfloor 1,1\rfloor, 0 + \lfloor 2,4\rfloor, \lfloor 3\rfloor + \lfloor 1,1\rfloor, \lfloor 3\rfloor + \lfloor 2,4\rfloor, \lfloor 2,4\rfloor + \lfloor 1,1\rfloor, \lfloor 2,4\rfloor + \lfloor 2,4\rfloor\rfloor \\ &= \lfloor\lfloor 1,1\rfloor,\lfloor 2,4\rfloor,\lfloor 3,1,1\rfloor,\lfloor 3,2,4\rfloor,\lfloor 2,4,1,1\rfloor,\lfloor 2,4,2,4\rfloor\rfloor \\ &= \alpha_1^2 + \alpha_2\alpha_4 + \alpha_1^2\alpha_3 + \alpha_2\alpha_3\alpha_4 + \alpha_1^2\alpha_2\alpha_4 + \alpha_2^2\alpha_4^2 \end{aligned}$$

which agrees with $(1 + \alpha_3 + \alpha_2\alpha_4)\left(\alpha_1^2 + \alpha_2\alpha_4\right) = \alpha_1^2 + \alpha_2^2\alpha_4^2 + \alpha_2\alpha_4 + \alpha_1^2\alpha_3 + \alpha_1^2\alpha_2\alpha_4 + \alpha_2\alpha_3\alpha_4$.

In general the **degree** of the multinumber M will be the maximum degree of its terms, and the **extent** of M will be the maximum extent of its terms.

Example 7 *For the multinumber* $A = \lfloor 0,_2 4, \lfloor 3,5 \rfloor, \lfloor 0,7,7,6 \rfloor \rfloor = 1 + 2\alpha_0^4 + \alpha_3\alpha_5 + \alpha_0\alpha_6\alpha_7^2$, *the degree of* A *is the degree of the term* $\alpha_0\alpha_6\alpha_7^2$, *which is* 4, *while the extent of* A *is the extent of that same term, which is* 7.

6 The caret operation

The successor operation to multiplication $\times = \times^{(1)}$is what we call the **caret operation,** and denote by $\wedge = \times^{(2)}$; for boxes $A, B, \cdots, K$ we define

$$A \wedge B \wedge \cdots \wedge K \equiv A \times^{(2)} B \times^{(2)} \cdots \times^{(2)} K \equiv \lfloor a \times b \times \cdots \times k : a \in A, b \in B, \cdots, k \in K \rfloor$$

the box of all possible products between elements of A and elements of B. This has some similarities with the usual exponentiation, but it is definitely different. In particular it is commutative and associative, and also distributes over addition for the same reason that multiplication does so that for example for any boxes A, B and C we have $A \wedge (B + C) = (A \wedge B) + (A \wedge C) = A \wedge B + A \wedge C$.

We can check that when we restrict to natural numbers, the caret operation reduces to just multiplication, so that for natural numbers m and n we have $m \wedge n = m \times n$. It follows that for any boxes A and B we have $s(A \wedge B) = s(A) \times s(B) = s(A) \wedge s(B)$.

If we consider the caret of monomials, we have $\lfloor m \rfloor \wedge \lfloor n \rfloor = \lfloor m \times n \rfloor$ which we can restate as $\alpha^m \wedge \alpha^n = \alpha^{m \times n}$ so this operation is distinct from multiplication on polynumbers.

Example 8 *Here is a caret of more general polynumbers:*

$$\lfloor 1,3,2 \rfloor \wedge \lfloor 2,4 \rfloor = \lfloor 1 \times 2, 1 \times 4, 3 \times 2, 3 \times 4, 2 \times 2, 2 \times 4 \rfloor = \lfloor 2,4,6,12,4,8 \rfloor$$

which in polynomial form is $\left(\alpha + \alpha^2 + \alpha^3\right) \wedge \left(\alpha^2 + \alpha^4\right) = \left(\alpha^2 + 2\alpha^4 + \alpha^6 + \alpha^8 + \alpha^{12}\right)$.

While the caret of polynumbers is seemingly curious, we shall see shortly that it plays a key role in fundamental number theory.

When we consider the caret operation acting on multinumbers, we discover something again a bit surprising: for natural numbers m and n we have

$$\alpha_m \wedge \alpha_n = \lfloor \lfloor m \rfloor \rfloor \wedge \lfloor \lfloor n \rfloor \rfloor = \lfloor \lfloor m \rfloor \times \lfloor n \rfloor \rfloor = \lfloor \lfloor m + n \rfloor \rfloor = \alpha_{m+n}.$$

Thus this arithmetical operation actually operates on the indices. This is reminiscent of the Umbral Calculus of G. C. Rota. Note that $\alpha_0 = \alpha$ becomes the caret identity, in the sense that for any box A we have

$$\alpha \wedge A = \lfloor 1 \rfloor \wedge A = A.$$

Example 9 *Here is a caret of more general multinumbers involving* $B = \lfloor 0, \lfloor 3 \rfloor, \lfloor 2,4 \rfloor \rfloor = 1 + \alpha_3 +$

$\alpha_2\alpha_4$ *and* $C = \lfloor\lfloor 1,1\rfloor, \lfloor 2,4\rfloor\rfloor = \alpha_1^2 + \alpha_2\alpha_4$*:*

$$\begin{aligned} B \wedge C &= \lfloor 0, \lfloor 3\rfloor, \lfloor 2,4\rfloor\rfloor \wedge \lfloor\lfloor 1,1\rfloor, \lfloor 2,4\rfloor\rfloor \\ &= \lfloor 0\times\lfloor 1,1\rfloor, 0\times\lfloor 2,4\rfloor, \lfloor 3\rfloor\times\lfloor 1,1\rfloor, \lfloor 3\rfloor\times\lfloor 2,4\rfloor, \lfloor 2,4\rfloor\times\lfloor 1,1\rfloor, \lfloor 2,4\rfloor\times\lfloor 2,4\rfloor\rfloor \\ &= \lfloor 0, 0, \lfloor 4,4\rfloor, \lfloor 5,7\rfloor, \lfloor 3,3,5,5\rfloor, \lfloor 4,6,6,8\rfloor\rfloor \\ &= 2 + \alpha_4^2 + \alpha_5\alpha_7 + \alpha_3^2\alpha_5^2 + \alpha_4\alpha_6^2\alpha_8. \end{aligned}$$

6.1 Sums of boxes and the caret product identity

There is another natural operator on boxes, which is the **summation operator** Σ, defined as follows: if $M = \lfloor A, B, \cdots, K\rfloor$ is a box, then the **summation of** M, denoted $\Sigma(M)$ is

$$\Sigma(M) \equiv A + B + \cdots + K.$$

Clearly if m is a natural number, then $\Sigma(m) = 0$. If p is a polynumber, say $p = p_0 + p_1\alpha + \cdots + p_k\alpha^k = \lfloor_{p_0} 0, _{p_1} 1, \cdots, _{p_k} k\rfloor$ then $\Sigma(p)$ is a natural number given by

$$\Sigma(p) = p_0 \times 0 + p_1 \times 1 + p_2 \times 2 + \cdots + p_k \times k = D(p)(1)$$

where $D(p) \equiv p_1 + 2p_2\alpha + \cdots + kp_k\alpha^{k-1}$ is the **derivative** of p. One can check that then for polynumbers p and q, we have

$$\Sigma(p \times q) = s(q) \times \Sigma(p) + s(p) \times \Sigma(q).$$

Example 10 *If* $p = \lfloor 3,3,4\rfloor$ *and* $q = \lfloor 1,2\rfloor$ *we have* $p \times q = \lfloor 3+1, 3+2, 3+1, 3+2, 4+1, 4+2\rfloor = \lfloor 4,5,4,5,5,6\rfloor$ *so that* $\Sigma(p\times q) = 29$ *while* $s(q)\times\Sigma(p) + s(p)\times\Sigma(q) = 2\times 10 + 3\times 3 = 29$.

The situation is somewhat cleaner for multinumbers and the caret operation. If r is a multinumber, say $r = \lfloor r_0, r_1, \cdots, r_l\rfloor$ for polynumbers $r_0, r_1, \cdots, r_l$ then $\Sigma(r) = r_0 + r_1 + \cdots + r_l$ is a polynumber. One can check that for multinumbers r and s we have

$$\Sigma(r \wedge s) = \Sigma(r) \times \Sigma(s)$$

and more generally for multinumbers $r, s, \cdots, w$ we have the **Caret product identity**:

$$\Sigma(r \wedge s \wedge \cdots \wedge w) = \Sigma(r) \times \Sigma(s) \times \cdots \times \Sigma(w).$$

Example 11 *Here is a caret of more general multinumbers:*

$$\lfloor 1,3,2\rfloor \wedge \lfloor 2,4\rfloor = \lfloor 1\times 2, 1\times 4, 3\times 2, 3\times 4, 2\times 2, 2\times 4\rfloor = \lfloor 2,4,6,12,4,8\rfloor$$

which in polynomial form is $(\alpha + \alpha^2 + \alpha^3)\wedge(\alpha^2 + \alpha^4) = (\alpha^2 + 2\alpha^4 + \alpha^6 + \alpha^8 + \alpha^{12})$. *Then* $\Sigma(\lfloor 1,3,2\rfloor) = 1 + 3 + 2 = 6$

while $\Sigma(\lfloor 2,4\rfloor) = 2+4 = 6$. Now $\Sigma(\lfloor 1,3,2\rfloor \wedge \lfloor 2,4\rfloor) = \Sigma(\lfloor 2,4,6,12,4,8\rfloor) = 2+4+6+12+4+8 = 36$.

Example 12 *This will play an important role in reformulating certain classical analytic number theory identities into statements in (finite) box arithmetic.*

For some of the further story, we will anticipate the extension of our basic number system from natural numbers to include also integers. An integer may be introduced as a formal difference $m - n$ of natural numbers m and n, with the equivalence $m - n \equiv m\prime - n\prime$ precisely when $m + n\prime = m\prime + n$. Addition and multiplication of integers follow from those of natural numbers in the usual way, and every natural number k embeds as the integer $k - 0$. With this extension, polynumbers may now carry integer coefficients, and the caret product of two polynumbers with integer coefficients is again a polynumber with integer coefficients. When we get to applications we will also use rational numbers, obtained in a similar formal fashion from integers by the usual formal quotients subject to an equivalence relation.

7 Caret forms of Dirichlet series from number theory

Let us give an application of how the box arithmetic developed so far allows us to revaluate some fundamentals of number theory, especially leaning on the caret operation to formulate a Fundamental Identity of Arithmetic involving prime factorizations and a corresponding algebra of Dirichlet series, now removed from the analytic context and requiring only finite computable arithmetic.

To set up Dirichlet algebra in box arithmetic, we first define the subtype of Poly which we call *-Poly: a polynumber p is a ***-polynumber** (read "star polynumber") precisely when it has no 0 elements. Then *-polynumbers form an associative commutative algebra under the operations of addition $+$ and caret $\wedge$, with identity $\lfloor 1 \rfloor$. The *-polynumbers up to degree n for a natural number n form a linear space *-Poly$_n$ with a basis the set $\{\lfloor 1 \rfloor, \lfloor 2 \rfloor, \cdots, \lfloor n \rfloor\}$. In the polynumber context, we introduced the notation $\alpha = \alpha^1, \alpha^2, \cdots, \alpha^n$ for these basis elements. This terminology is well suited for the usual algebra structure on polynumbers, but not for the caret algebra *-Poly on *-polynumbers. Instead we introduce the **Dirichlet form** for these boxes:

$$1^s \equiv \lfloor 1 \rfloor, 2^s \equiv \lfloor 2 \rfloor, \cdots, n^s \equiv \lfloor n \rfloor$$

where now s is just part of the expression and does not signify (at this stage!) an exponentiation. Classically the reciprocal $1/n^s$ is used to ensure convergence of the series, but in our finite algebraic setting no such requirement exists, so n^s is the natural form. Then the rule for the caret product $\lfloor m \rfloor \wedge \lfloor n \rfloor = \lfloor m \times n \rfloor$ becomes

$$m^s \wedge n^s = (m \times n)^s .$$

Happily this accords with our usual conventions for exponents, and so is easy to remember. In the usual formulation the reciprocal is used instead, so $\frac{1}{n^s}$ instead of n^s, in order to facilitate convergence of certain infinite sums that appear, but we have no need of this requirement.

So to summarize: for polynumbers the basis element $\lfloor n \rfloor$ is given the form α^n for an arbitrary choice of the (Greek) letter α, while for *-polynumbers the same basis element $\lfloor n \rfloor$ is given the form n^s for an arbitrary choice of the letter s. The consequence is that in both cases the resulting algebra multiplication which is $\times$ for polynumbers and $\wedge$ for *-polynumbers aligns smoothly with our familiar algebra.

We are now in a position to introduce many of the fundamental sequences in number theory, all expressed in the language of *-polynumbers and with multiplication the Dirichlet product which turns out to be exactly the caret operation. We introduce them with their usual names but with our somewhat novel notation.

The simplest case is the **delta $*-$polynumber** is $\delta \equiv 1^s = \lfloor 1 \rfloor$. This is the multiplicative identity in the caret algebra *-Poly. Many other examples come in series indexed by a natural number $n \geq 1$, and are derived from multiplicative functions from natural numbers to natural numbers. Arguably the most important is the **zeta $*-$polynumber**

$$\zeta_n \equiv \sum_{1 \leq k \leq n} k^s = 1^s + 2^s + \cdots + n^s = \lfloor 1, 2, \cdots, n \rfloor .$$

The **identity series** is

$$N_n \equiv \sum_{1 \leq k \leq n} k \times k^s = 1 \times 1^s + 2 \times 2^s + 3 \times 3^s + \cdots + n \times n^s.$$

The **Euler totient series** is

$$\varphi_n \equiv \sum_{1 \leq k \leq n} \varphi(k) \times k^s = 1^s + 2^s + 2 \times 3^s + 2 \times 4^s + 4 \times 5^s + 2 \times 6^s + 6 \times 7^s + \cdots + \varphi(n) n^s$$

where $\varphi(k)$ is the number of natural numbers less than k relatively prime to it. The **divisor count series** is

$$d_n \equiv \sum_{1 \leq k \leq n} d(k) \times k^s = 1^s + 2 \times 2^s + 2 \times 3^s + 3 \times 4^s + 2 \times 5^s + 4 \times 6^s + 2 \times 7^s + \cdots + d(n) n^s$$

where $d(k)$ is the number of divisors of k. The **divisor sum series** is

$$\sigma_n \equiv \sum_{1 \leq k \leq n} \sigma(k) \times k^s = 1^s + 3 \times 2^s + 4 \times 3^s + 7 \times 4^s + 6 \times 5^s + 12 \times 6^s + 8 \times 7^s + \cdots + \sigma(n) n^s$$

where $\sigma(k)$ is the sum of the divisors of k. The **square zeta series** is

$$S_n \equiv \sum_{1 \leq k \leq n} S(k) \times k^s = 1^s + 4^s + 9^s + \cdots \left(n^2\right)^s$$

where $S(k)$ is 1 when k is a square and is 0 otherwise. And of course there are many others, for multiplicative functions can be multiplied and raised to powers to generally also create more multiplicative functions. In particular, once we have augmented our number system to also include integers, then there are additional interesting series. The **Mobius $*-$polynumber** is

$$\mu_n \equiv \sum_{1 \leq k \leq n} \mu(k) \times k^s = 1^s - 2^s - 3^s - 5^s + 6^s - 7^s \pm \cdots + \mu(n) n^s$$

where $\mu(n) = (-1)^k$ if n has k distinct prime factors, and 0 otherwise. The **Liouville series** is

$$\lambda_n \equiv \sum_{1\leq k\leq n} \lambda(k) \times k^s = 1^s - 2^s - 3^s + 4^s - 5^s + 6^s - 7^s \pm \cdots \pm \lambda(n)\, n^s$$

where $\lambda(n) = (-1)^{\Omega(n)}$ where $\Omega(n)$ is the number of prime factors of n including multiplicity.

Now there are many often remarkable algebraic relations between these various series, which are efficiently written using the caret operation, suitably truncated. Here are some: for any natural number n,

$$\zeta_n \wedge \zeta_n =_n d_n \qquad \zeta_n \wedge N_n =_n \sigma_n \qquad \varphi_n \wedge \zeta_n =_n N_n \qquad \mu_n \wedge \zeta_n =_n \delta \qquad \lambda_n \wedge \zeta_n = S_n.$$

The first says that (informally) the Dirichlet square of the zeta series is the divisor count series; the second is that the zeta series convolved with the identity series gives the divisor sum series, and so on. We illustrate each of these with $n = 6$, the first three are over the natural numbers and we express them using box form, while the remaining two require extension to integers and are expressed in Dirichlet form:

$$\begin{aligned}
\zeta_6 \wedge \zeta_6 &= \lfloor 1,2,3,4,5,6 \rfloor \wedge \lfloor 1,2,3,4,5,6 \rfloor =_6 \lfloor 1,_2 2,_2 3,_3 4,_2 5,_4 6 \rfloor =_6 d_6 \\
\zeta_6 \wedge N_6 &= \lfloor 1,2,3,4,5,6 \rfloor \wedge_6 \lfloor 1,_2 2,_3 3,_4 4,_5 5,_6 6 \rfloor =_6 \lfloor 1,_3 2,_4 3,_7 4,_6 5,_{12} 6 \rfloor =_6 \sigma_6 \\
\varphi_6 \wedge \zeta_6 &= \lfloor 1, 2,_2 3,_2 4,_4 5,_2 6 \rfloor \wedge \lfloor 1,2,3,4,5,6 \rfloor =_6 \lfloor 1,_2 2,_3 3,_4 4,_5 5,_6 6 \rfloor =_6 N_6 \\
\mu_6 \wedge \zeta_6 &= (1^s - 2^s - 3^s - 5^s + 6^s) \wedge (1^s + 2^s + 3^s + 4^s + 5^s + 6^s) =_6 1^s = \delta \\
\lambda_6 \wedge \zeta_6 &= (1^s - 2^s - 3^s + 4^s - 5^s + 6^s) \wedge (1^s + 2^s + 3^s + 4^s + 5^s + 6^s) =_6 1^s + 4^s + 5^s = S_6.
\end{aligned}$$

So now in fact we can given meaning to all these assertions with seemingly unbounded series, which are now to be interpreted as consistent families of finite *-polynumbers indexed by a natural number n: If the understanding is clear, we could introduce the notation $f \wedge g = h$ to mean that for any natural number n, we have $f_n \wedge g_n = h_n$: for this no analysis or notion of convergence is necessary. Indeed the ideal limiting object is just a short form for a measured family of completely explicit objects, indexed by a natural number n.

8 The Fundamental Identity of Arithmetic

For any prime number p and any natural number $1 \leq n$ let $m(p,n)$ be the largest natural number such that $p^m \leq n$. Then for a prime p and a natural number n, define the **prime power *-polynumber**

$$\rho(p)_n \equiv \sum_{0\leq k\leq m(p,n)} \left(p^k\right)^s = \left\lfloor 1, p, p^2, \cdots, p^{m(p,n)} \right\rfloor = 1^s + (p)^s + \left(p^2\right)^s + \cdots + \left(p^{m(p,n)}\right)^s.$$

Then the degree of $\rho(p)_n$ is less than or equal to n.

Example 13 *If* $p = 5$ *and* $n = 130$ *then* $p(5)_{130} = \lfloor 1, 5, 25, 125 \rfloor = 1^s + 5^s + 25^s + 125^s$.

Here then is a precise version of the informal identity from the Introduction.

Theorem 14 (Fundamental Identity of Arithmetic (FIA)) *Suppose that n is a natural number and that we have, in order, all of the prime numbers $2, 3, 5, \cdots, p$ less than or equal to n. Then*

$$\rho(2)_n \wedge \rho(3)_n \wedge \cdots \wedge \rho(p)_n =_n \zeta_n.$$

Explicitly this is the identity

$$\left\lfloor 1, 2, 2^2, \cdots, 2^{m(2,n)} \right\rfloor \wedge \left\lfloor 1, 3, 3^2, \cdots, 3^{m(3,n)} \right\rfloor \wedge \cdots \wedge \left\lfloor 1, p, p^2, \cdots, p^{m(p,n)} \right\rfloor =_n \lfloor 1, 2, 3, \cdots, n \rfloor$$

*a caret product over all the prime power *-polynumbers $\rho(q)_n$ where q runs from 2 to p.*

This is really just a combinatorial restatement of the usual Fundamental Theorem of Arithmetic that every natural number from 1 to n is uniquely the product of powers of the primes $2, 3, \cdots, p$, but now it is in the form of *an equality of boxes.*

Example 15 *If $n = 12$ then the primes less than or equal to 12 are $2, 3, 5, 7, 11$, and the identity is* $\lfloor 1, 2, 4, 8 \rfloor \wedge \lfloor 1, 3, 9 \rfloor \wedge \lfloor 1, 5 \rfloor \wedge \lfloor 1, 7 \rfloor \wedge \lfloor 1, 11 \rfloor =_{12} \lfloor 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 \rfloor$.

Note that the Fundamental Identity is maintained if we uniformly apply a multiplicative function to natural numbers.

Example 16 *By applying the squaring operator for example, we may now deduce the identity* $\lfloor 1, 4, 16, 64 \rfloor \wedge \lfloor 1, 9, 81 \rfloor \wedge \lfloor 1, 25 \rfloor \wedge \lfloor 1, 49 \rfloor \wedge \lfloor 1, 121 \rfloor =_{144} \lfloor 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144 \rfloor$ *where the truncation level has also been squared.*

Or we could apply a number theoretic function, such as the number of divisors $d(n)$ or the sum of divisors $\sigma(n)$ or even some power of these. If we work over the rational numbers, and apply the multiplicative function which takes a natural number n and raises it to some negative power, for example to get $\frac{1}{n^3}$, then there is the possibility that we could apply the caret product identity to make some statement about limiting values of summations as our truncations get larger and larger.

8.1 Calculating a limiting summation interval

To illustrate this just with an example, suppose we are interested in an approximate value for the sum

$$S(N) = \sum_{0<n\leq N} \frac{1}{\sigma(n)^3}$$

for a large natural number N. Our aim is to use the Box Arithmetic framework to allow us to subtly shift from answering such questions with classical analysis involving real numbers and infinite operations, to purely finite algebraic computations, generally depending on some parameters.

So to make this question more precise, we could ask instead for a rational number interval $[t, u]$ such that for all sufficiently large natural numbers N, we have $t \leq S(N) \leq u$, and such could be called an **eventual bounding interval** for the sums $S(N)$. Note that a priori such an eventual bounding interval need not exist, although of course a lower bound t is readily found since the terms are all positive: just take $t = S(M)$ for any natural number M, or even $t = 0$. To make things specific, lets try to find t and u such that $u - t \leq 1/10$.

We shall restrict the FIA just to the primes $2, 3, 5$ and 7, and only take powers up to 2. Then the caret product $M \equiv \lfloor 1, 2, 2^2 \rfloor \wedge \lfloor 1, 3, 3^2 \rfloor \wedge \lfloor 1, 5, 5^2 \rfloor \wedge \lfloor 1, 7, 7^2 \rfloor$ is a box of 81 distinct natural numbers (just those which are either 1 or have prime factorizations involving only the primes $2, 3, 5$ and 7 with exponents only up to 2). In fact this box is

$$M = \left\lfloor \begin{array}{c} 1, 2, 3, 4, 5, 6, 7, 9, 10, 12, 14, 15, 18, 20, 21, 25, 28, 30, 35, 36, 42, 45, 49, 50, 60, 63, 70, 75, 84, \\ 90, 98, 100, 105, 126, 140, 147, 150, 175, 180, 196, 210, 225, 245, 252, 294, 300, 315, 350, 420, \\ 441, 450, 490, 525, 588, 630, 700, 735, 882, 900, 980, 1050, 1225, 1260, 1470, 1575, 1764, 2100, \\ 2205, 2450, 2940, 3150, 3675, 4410, 4900, 6300, 7350, 8820, 11025, 14700, 22050, 44100. \end{array} \right\rfloor .$$

For a prime p, if we apply the transformation Ψ that sends n to $\frac{1}{\sigma(n)^3}$ to the box $\lfloor 1, p, p^2 \rfloor$ we get the box $R(p) = \left\lfloor 1, \frac{1}{\sigma(p)^3}, \frac{1}{\sigma(p^2)^3} \right\rfloor$ with sum $f(p) \equiv \Sigma(R(p)) = 1 + \frac{1}{(1+p)^3} + \frac{1}{(1+p+p^2)^3}$. Then because of the multiplicative nature of the transformation Ψ, the Caret product identity applies and gives

$$t \equiv f(2) \times f(3) \times f(5) \times f(7) = \frac{281\,703\,168\,422\,010\,537\,266\,399}{264\,837\,830\,174\,132\,729\,020\,416} = \sum_{m \in M} \frac{1}{\sigma(m)^3}$$

and this rational number t is just approximately 1.06368. So t is clearly a lower bound for $S(N)$ provided that $N \geq 44100$.

An elementary upper bound for $S(N)$ may be obtained by observing that for any natural number n we have $\sigma(n) \geq 1+n$ so that $\frac{1}{\sigma(n)^3} \leq \frac{1}{(n+1)^3}$ and so $S(N) = 1 + \sum_{2 \leq n \leq N} \frac{1}{\sigma(n)^3} \leq 1 + \sum_{2 \leq n \leq N} \frac{1}{(n+1)^3}$. Now since $1/x^3$ is decreasing, for any natural number k we have $\frac{1}{(k+1)^3} \leq \int_k^{k+1} \frac{1}{x^3} dx$ and so

$$S(N) \leq 1 + \sum_{2 \leq n \leq N} \frac{1}{(n+1)^3} \leq 1 + \int_2^\infty \frac{1}{x^3} dx = \frac{9}{8}$$

where this last integral can be defined and computed purely within the rational number framework as it has a rational value. We conclude that for $N \geq 44100$ we have

$$S(N) \in \left[\frac{281\,703\,168\,422\,010\,537\,266\,399}{264\,837\,830\,174\,132\,729\,020\,416}, \frac{9}{8} \right] \simeq \quad 1.0637 \quad 1.125$$

of width approximately 0.061 which is indeed less than our desired goal of $1/10$. If we use a computer to evaluate the sum for larger N, we get something close to $S(N) \simeq 1.0656$, reflecting the fact that the 81-element partial Euler product captures most of the sum. The upper bound $9/8$ is elementary and explicit. A sharper upper bound can be obtained by computing finitely many additional σ values, but our point here is methodological: rigorous rational intervals replace real-number limits, with precision improvable on demand by purely finite computation.